

BayeSN: Scalable Hierarchical Modelling of Type Ia Supernovae

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$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$

- Tripp formula - one parameter for two separate effects
- Correctly handling dust is key for both SN Ia astrophysics and cosmology
- If intrinsic effect misattributed to dust, could lead to bias

- For typical Bayesian inference, each supernova would be fit separately

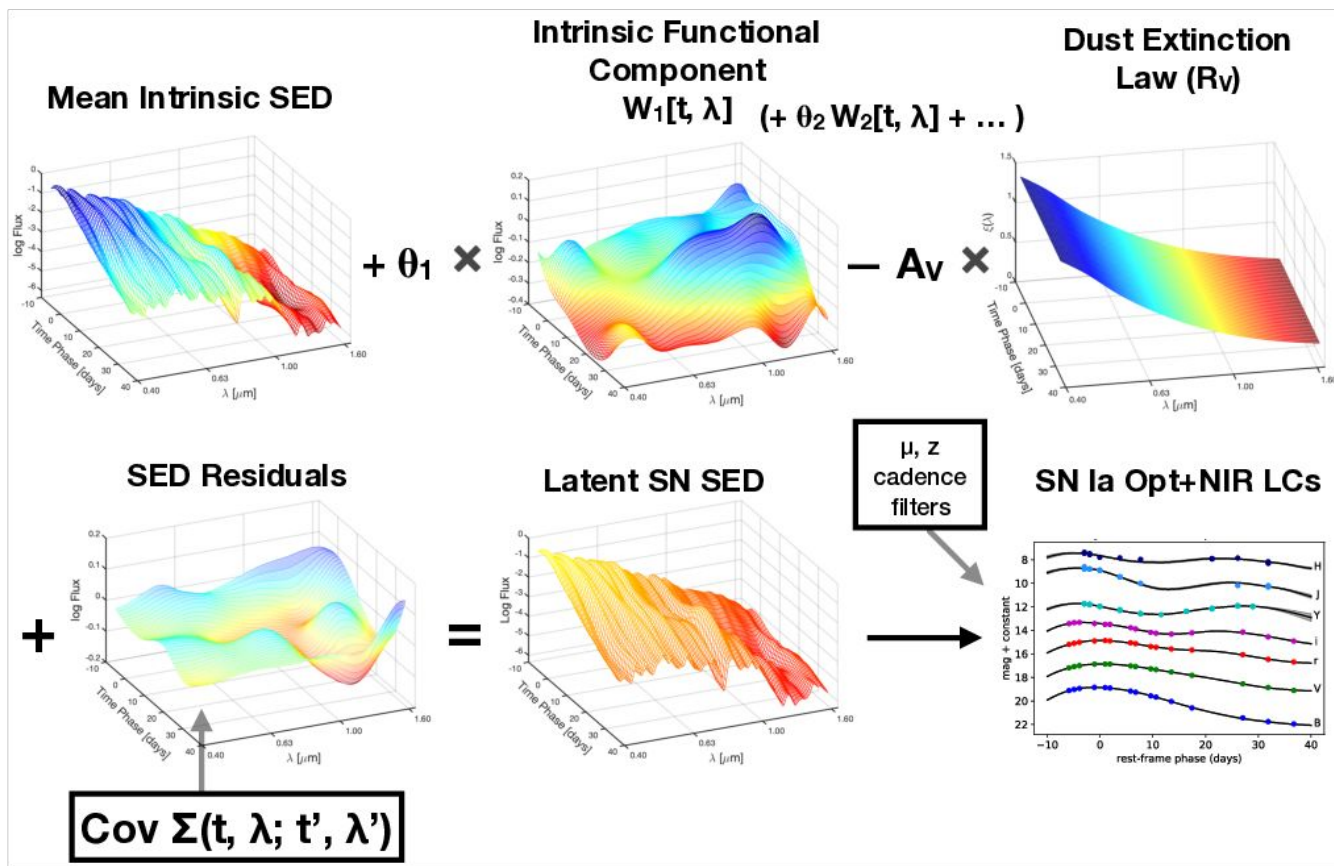
$$R_V \sim U(1, 5)$$

- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

- We can model intrinsic variation and dust properties at the population level as two separate effects

The BayeSN Model



- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation
 - Better estimation of global properties
- Full SED model - all colour/magnitude information used when estimating distance rather than just single colour/magnitude pairs
- Model extends into NIR wavelengths - better SN Ia standardisation and constraint of host galaxy dust

- Lots of parameters (e.g. ~ 400 global, ~ 4200 latent parameters for Thorp 2021)
- Complex likelihood evaluation:
 - Compute SED model
 - Evaluate numerical integrals through different filters, for each time of observation and each supernova
- Computationally expensive

Problem:

Scaling BayeSN for next generation data sets without compromising functionality

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Solution:

pyro-ppl/**numpyro**



Probabilistic programming with NumPy powered by
JAX for autograd and JIT compilation to
GPU/TPU/CPU.

 82
Contributors

 529
Used by

 2k
Stars

 179
Forks

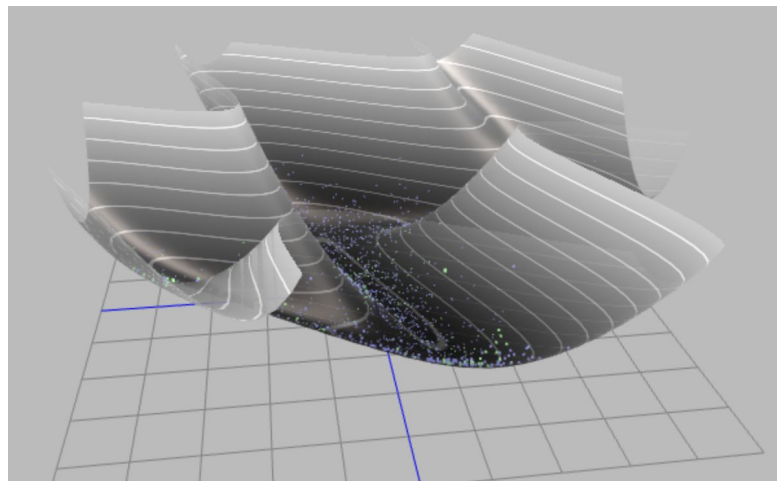


Jax

- Numpy with JIT compilation for GPUs
- Supports autodiff (derivatives for free), ideal for Hamiltonian Monte Carlo
- Very efficient matrix/tensor operations

Numpyro

- Probabilistic programming package for Python built on Jax



Credit: Alex Rogozhnikov

Vectorized likelihood evaluation + GPUs = Fast Bayesian inference

Benchmarks

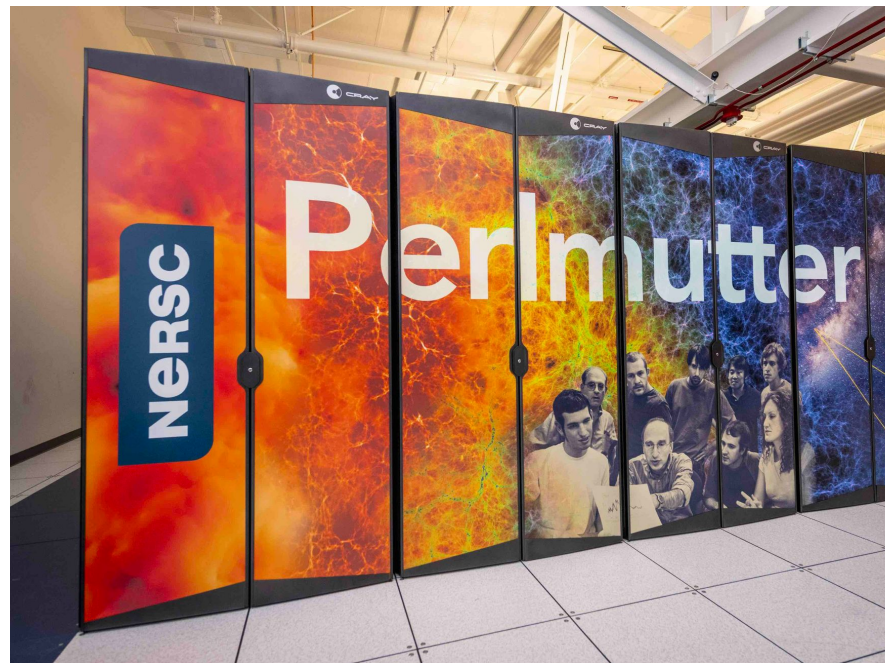
On Perlmutter, using 4 GPUs

Thorp+21:

- 157 *griz* Foundation light curves
- Training ~ 1 day $\rightarrow \sim 10$ minutes
- Fits ~ 2 minutes per object $\rightarrow \sim 4$ minutes for all 157

Mandel+20:

- 79 optical+NIR light curves
- Training ~ 5 days $\rightarrow \sim 20$ minutes
- Fits ~ 40 minutes per object $\rightarrow \sim 18$ minutes for all 79

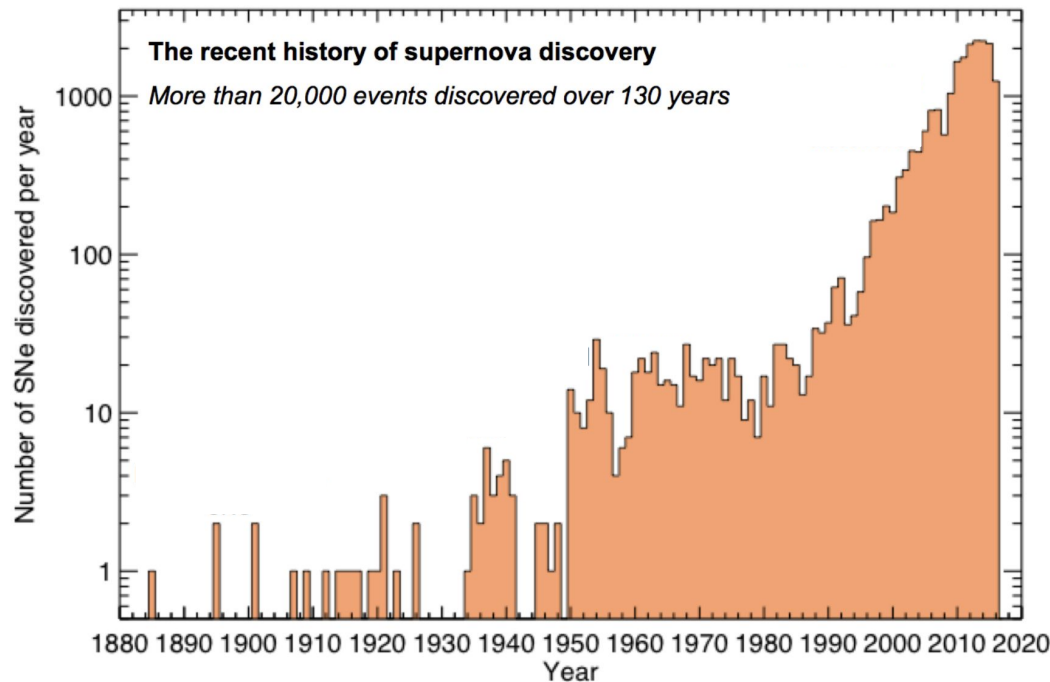


Credit: NERSC

Testing scalability with simulated
Foundation-like SNe:

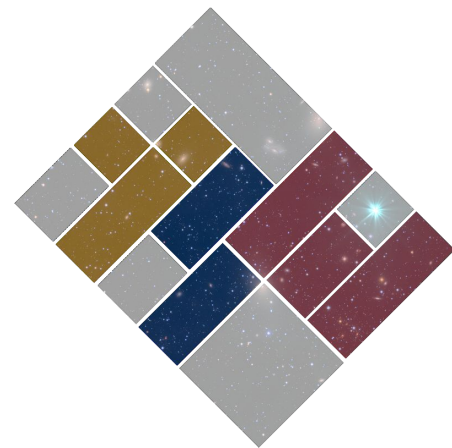
- 100 SNe
 - Training - 9 mins
 - Fitting - 3 mins
- 1000 SNe
 - Training - 21 mins
 - Fitting - 18 mins

Efficiency of tensor operations
improves as sample size grows,
model is scalable by construction



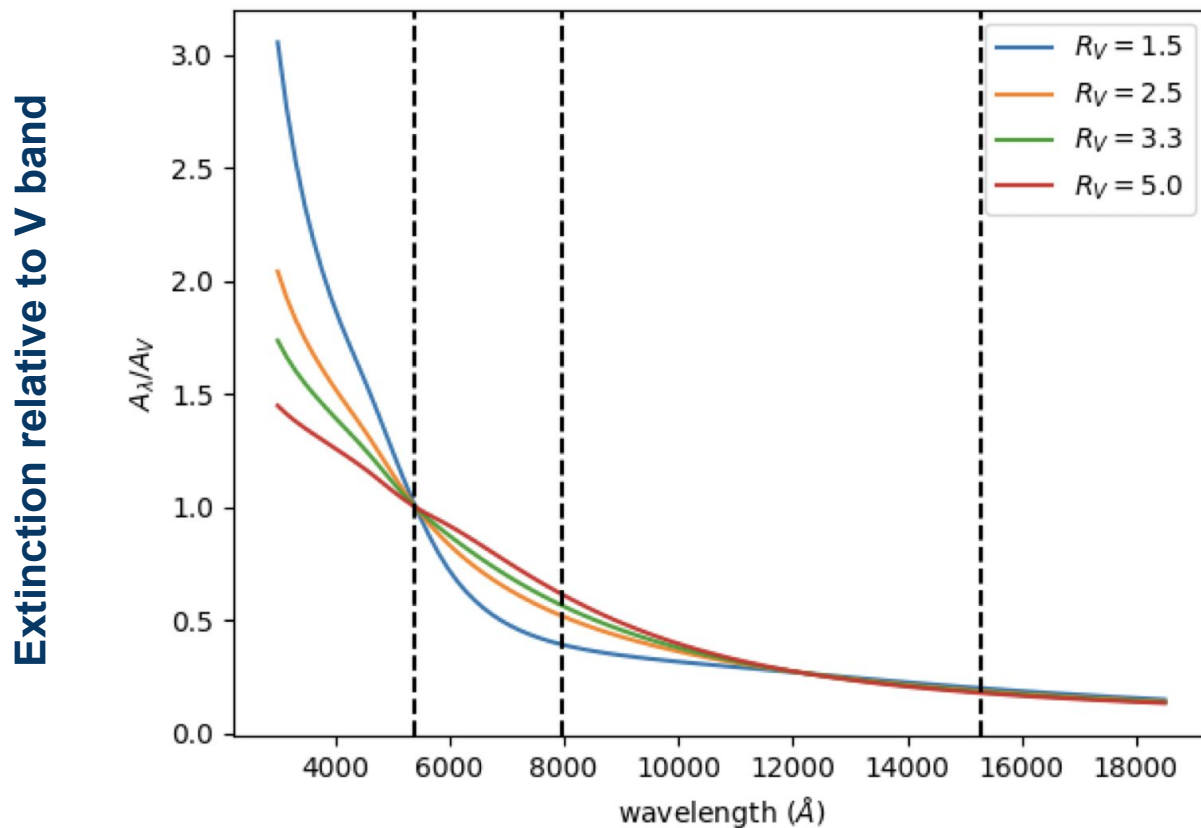
Credit: Mark Sullivan

- Previously, training and fitting on data-sets $> 10,000$ SNe would have been computationally unfeasible - now achievable in relatively short timescales
- Work ongoing to implement BayeSN within SNANA for cosmological analyses
- Planned hierarchical dust analysis of hundreds of SNe Ia from the Young Supernova Experiment (YSE), as well as archival samples
- Ideal framework to look at dust and intrinsic distributions as a function of environment



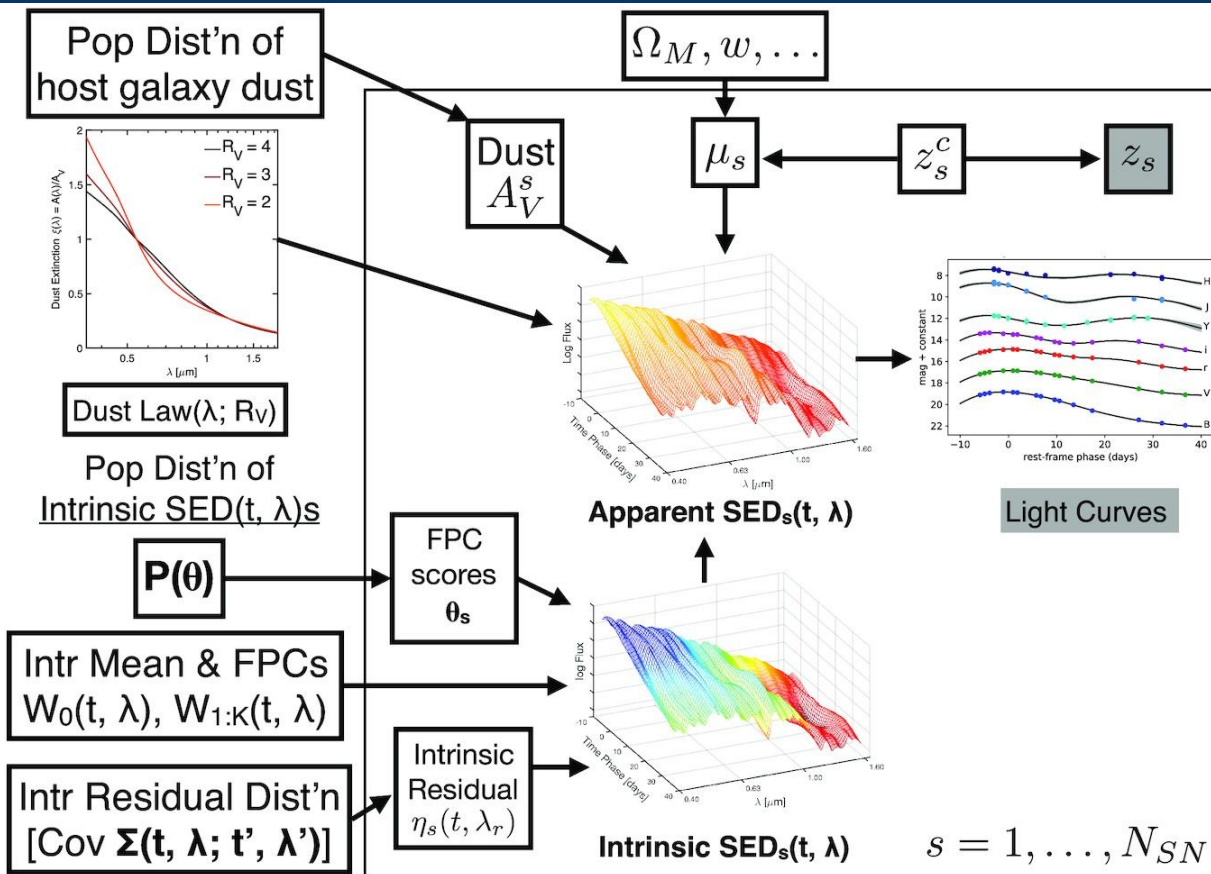
- BayeSN is a hierarchical model for fitting type Ia supernovae with several advantages over current approaches
- Through the use of jax/numpyro and GPU acceleration, we have been able to achieve $\sim 100\times$ performance speed ups, making the use of BayeSN feasible for large surveys
- BayeSN being implemented in SNANA for cosmological analysis
- Lots of potential for analysis of dust and intrinsic distributions of SNe Ia

Extinction vs. wavelength



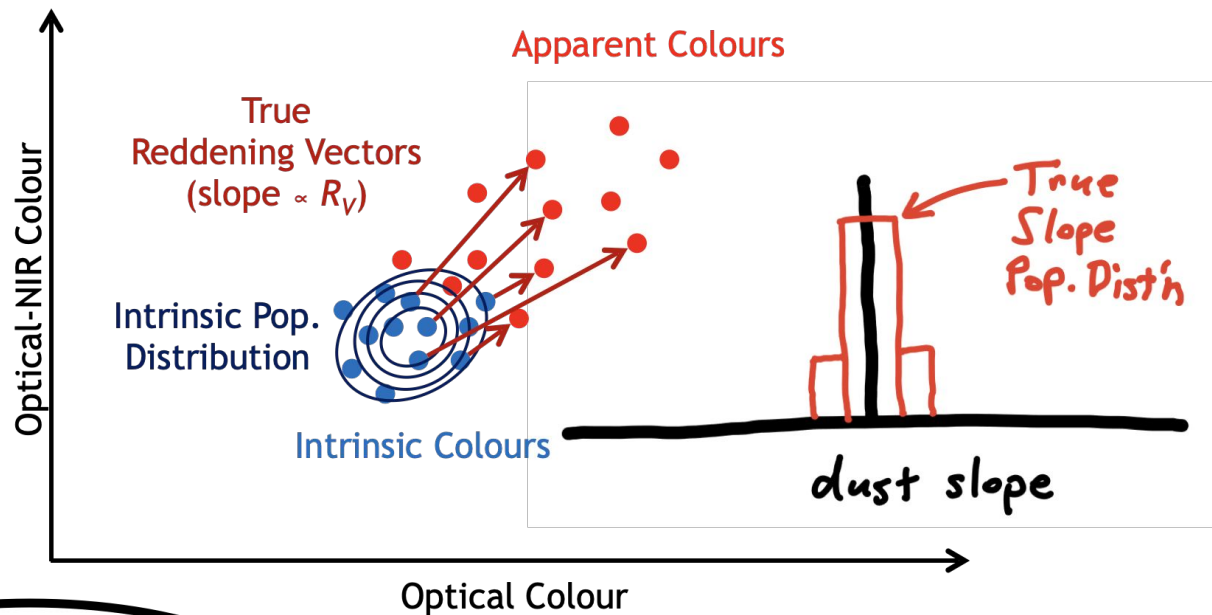
Lower $R_V \rightarrow$
steeper dust law

The BayeSN Model



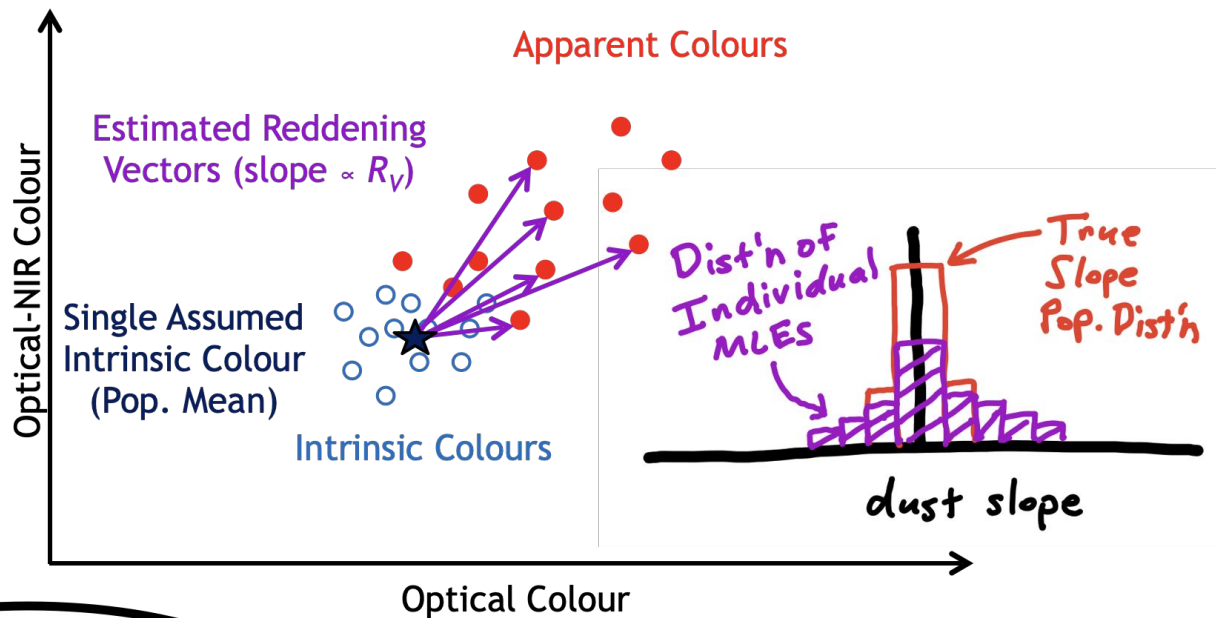
Mandel (2020)

Why include intrinsic variation?



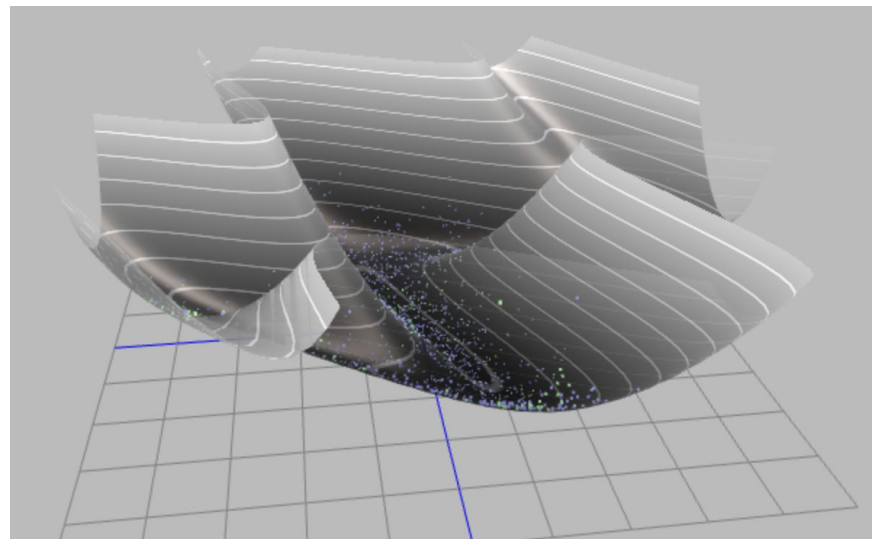
Need to account for
intrinsic colour scatter
when inferring R_V !

Why include intrinsic variation?



Otherwise width of R_V
distribution will be
overestimated!

- Two modes of use
 - Training - Conditioning global parameter values on data
 - Fitting - Inference of supernova properties with fixed global parameters
- Uses Hamiltonian Monte Carlo (HMC) to sample posteriors
 - MCMC method using gradient to make efficient steps



Credit: Alex Rogozhnikov

- Light curves are not of a fixed shape, but we want vectorized calculations
- For a given data set, all light curves are padded with zeros to match the longest time series. Numpyro supports masking - padded data points do not contribute to likelihood
- Allows all flux integrals across all SNe, phases and bands to be calculated as a single tensor operation - calculations scale very efficiently
- Using HMC, we can fit multiple SNe simultaneously with shared flux integral calculations